DOCUMENT 1 OF 4 · PROTOCOL (WRITTEN BEFORE THE CONFIRMATORY RUNS)

Self-Control in Two Cognitive Architectures

Task, Framework, Hypotheses and Simulation Protocol

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What this document is

This is the protocol for a study comparing how two cognitive architectures, CLARION and Soar, produce self-control in the same robot task. It defines the task, the self-control framework, the three models to be compared, what every simulation must implement, the measures, the hypotheses, the ablations and the analysis plan.

It was written *before* the confirmatory simulations were run. The models were built and debugged on development seeds (5001 and up), which are never used for hypothesis tests. The confirmatory seeds (1–200) were not run before this version was frozen. Any later change is reported as a deviation (Section 12). Document 2 describes the CLARION simulation, Document 3 the Soar simulations, and Document 4 the comparison in Webots.

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1 Purpose and research questions

Gomez and Sun (2024) proposed a "rational" framework in which self-control is not a fixed amount of willpower, but a metacognitive decision about *how explicit to be*. Explicit processing has a benefit (it serves the drives behind the current goal) and a cost. The framework, implemented in the CLARION cognitive architecture (Sun, 2016), reproduced three human findings: counteractive control when a choice is ahead (Myrseth et al., 2009; Gomez and Sun, 2025), the decline of control after a demanding task (Schmeichel, 2007), and the moderation of implicit and explicit attitudes by cognitive capacity (Friese et al., 2008). A Python reconstruction (pyclarion-sc) reproduces these simulations and carries them over to a cleaning robot.

The study asks three questions.

RQ1. Do the framework's three effects (anticipation, motivational shift, capacity) appear when CLARION controls an embodied agent, rather than a simulated participant?

RQ2. Which architectural commitments produce which effect? A native Soar model (Laird, 2012) built from Soar's own mechanisms (reinforcement learning, impasse-driven deliberation and chunking), with no drives and no cost of explicitness, gives a principled contrast.

RQ3. Is the framework a property of the theory or of one implementation? The same CLARION equations written as Soar rules should behave identically (an equivalence test).

The study is computational. It does not collect new human data. Its contribution is to identify which mechanisms are necessary for which self-control phenomenon, by comparing architectures on one task under controlled conditions, with ablations.

2 The self-control framework

2.1 Subsystems

CLARION has four subsystems; the framework uses three of them in the robot. The **motivational subsystem (MS)** holds drives. The **metacognitive subsystem (MCS)** sets goals and decides

how explicit processing should be. The **action-centered subsystem (ACS)** has a *top level* of explicit rules and a *bottom level* of implicit networks. Self-control conflicts arise when the two levels recommend different actions.

2.2 Equations

Equation numbers in brackets refer to Gomez and Sun (2024).

$$ds_d = stimulus_d \times deficit_d \qquad \text{drive strength [Eq. 5]} \qquad (1)$$

$$deficit_d(t) = \max(0,\ deficit_d(0) - k_d\, t) \qquad \text{deficit on a satisfying task [Eq. 6]} \qquad (2)$$

$$gs_g = \textstyle\sum_d relevance_{d\to g}\, ds_d \qquad \text{goal strength [Eq. 7]} \qquad (3)$$

$$value(g) = \textstyle\sum_d ds_d\, satisfaction_d(g) \qquad \text{benefit of the goal [Eq. 3]} \qquad (4)$$

$$U_j = j\,(\alpha\, value(g) - c), \quad j = 0, \ldots, 20 \qquad \text{utility of explicitness level } j \text{ [Eq. 2]} \qquad (5)$$

$$P(x) = \frac{e^{v_x/\tau}}{\sum_y e^{v_y/\tau}} \qquad \text{Boltzmann choice [Eq. 4]} \qquad (6)$$

The MCS draws a goal from the goal strengths and an explicitness level j from the utilities. With $p_{\text{explicit}} = j/20$, the top level selects the action if one of its rules applies; otherwise the bottom level's Q-values select it. The coding of levels as $0, \ldots, 20$ is a reconstruction choice, documented in pyclarion-sc (the papers say only that higher codes mean higher explicitness).

2.3 What the framework predicts

Because U_j is linear in j, the sign of the *net benefit* $\alpha\, value(g) - c$ decides whether high or low explicitness is favoured. Anything that raises the relevant drive raises control; anything that lowers it lowers control:

- **Anticipation.** A choice that is still ahead raises the stimulus of the achievement drive (0.8 vs. 0.5 after the choice), so control is stronger before the choice point.
- **Motivational shift.** Time on a drive-satisfying, control-demanding task lowers the deficit (Eq. 6), so control is weaker *afterwards*. The framework attributes the apparent "depletion" to motivation, not to a resource.
- **Capacity.** When processing capacity is limited, explicitness is constrained by the situation (0.4–0.6 instead of 0.7–1.0, following the Friese simulation), so the habit acts more often.

3 The task: a cleaning robot that passes a tempting room

3.1 World

The world is pyclarion-sc's grid world (Figure 1). Rooms A and B are assigned to the robot. Room C is not assigned, but it is very dirty (dirt 0.75–1.0 in every cell, against 0.25–0.75 in 70% of the assigned cells). A corridor connects the rooms and holds the charging dock. The robot sees dirt in its own room and, through an open door, cells within 4.5 cells of that door.

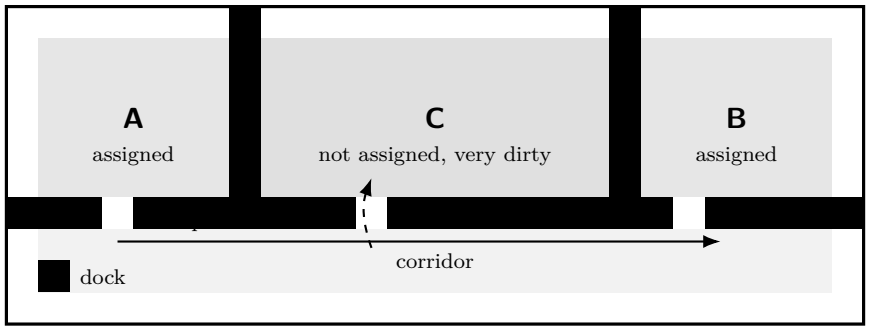


Figure 1: The grid world (27 × 10 cells). After room A the robot must pass the open door of room C on its way to room B. Walls are black; doors are the gaps.

The robot moves one cell per step and cleans the cell it is on. Battery use is 0.25% per move, 0.15% per cleaning step, and 0.1% for every explicit decision (the cost of planning). Below 8% battery a safety reflex sends the robot to the dock, outside the self-control mechanism. An episode ends when rooms A and B are clean and the robot is at the dock, when the battery is empty, or after 700 steps.

3.2 Behaviours

Every architecture chooses among the same five behaviours ("options"), each of which runs for up to three steps before the next decision: `clean_quiet` (low suction), `clean_full`, `next_room` (go to the next assigned room), `detour` (go to the most attractive visible dirt) and `dock`. A decision is also made early when a behaviour cannot continue.

3.3 Scenarios

Three scenarios carry the three human findings over to the robot. Each has two conditions.

Scenario	Human study	Manipulation in the robot	Conditions
Anticipation	Myrseth et al. (2009)	The choice point is room C's door. While it is ahead, the achievement stimulus is 0.8; after it, 0.5.	*anticipation* vs. *constant* (0.65 throughout)
Shift	Schmeichel (2007)	Room A is a quiet zone: the plan requires low suction there. Following the rule satisfies achievement (0.9), so its deficit falls with time ($k = 0.54$).	*quiet* vs. *free* (no quiet zone, satisfaction 0.6, $k = 0.36$)
Capacity	Friese et al. (2008)	After room A the robot rebuilds its map, a heavy job that lasts until room B is clean.	*busy* vs. *free* computer

The parameters of each scenario come from the corresponding simulation in Gomez and Sun (2024, 2025). The grid-world quantities (battery costs, sight radius, dirt levels) are chosen, not fitted.

4 Models to be compared

4.1 M1: Clarion (pyclarion-sc)

The robot controller of pyclarion-sc 0.5.0, unchanged. MS: achievement (stimulus × deficit, Eq. 5–6) and energy ($0.6 \times (1 - \text{battery})$). MCS: goals *clean assigned rooms* and *recharge* (Eq. 7, Boltzmann at $\tau = 0.1$), with goal persistence (the goal is kept unless another is stronger by more than 0.1); explicitness by Eq. 2–4 ($\alpha = 1$, $c = 0.185$, $\tau = 0.7$; shift scenario $c = 0.16$, $\tau = 0.8$). ACS top level: five plan rules (P1–P5). ACS bottom level: an MLP (6-16-16-5) trained on a greedy, dirt-chasing habit, Boltzmann at $\tau = 0.08$.

4.2 M2: Clarion-in-Soar (implementation check)

The same model written as 64 Soar productions (Soar 9.6.5). Drive strengths, goal strengths and $value(g)$ are i-supported elaborations; every Boltzmann choice becomes a set of operators with numeric-indifferent preferences equal to the CLARION value divided by its temperature, selected by Soar's Boltzmann rule at temperature 1, which reproduces CLARION's probabilities exactly. The bottom level uses the habit function that M1's network was trained to imitate, written as rules. M2 is *not* an architectural contrast. It tests whether the predictions belong to the theory rather than to one code base.

4.3 M3: native Soar

A model built only from Soar's own mechanisms, with no drives, no explicitness utility and no cost term (45 productions plus the learned habit).

- **Habit: Soar-RL.** Numeric preferences for each (situation, behaviour) pair, learned with reward = dirt cleaned in any room. Situations are discretized: in an assigned room with dirt; quiet zone; temptation none / low (< 0.5) / high; temptation near (≤ 5 cells) / far; battery low (< 0.25) / mid (< 0.6) / high; assigned rooms done. Trained once for 300 episodes (seeds 10001–10300; learning rate 0.1, discount 0.9, temperature 0.3), then frozen (temperature 0.1 at test).
- **Temptation is proposed.** Every behaviour the world affords is proposed, including `detour`.
- **Conflict.** Explicit task knowledge marks options that are off-task (`detour`; docking with work left and battery not low), break a constraint (full suction in a quiet zone) or are risky (anything but docking on low battery).
- **Control by deliberation.** A conflict with no control knowledge yet triggers the operator `deliberate`. It has no application, so Soar creates an operator no-change substate. There each candidate is evaluated against the task, one decision cycle per candidate, and a summary is returned to the top state as control knowledge (reject the off-task options, prefer the best one).
- **Capacity as time.** Deliberation takes decision cycles. In the capacity scenario each robot cycle has a budget of decision cycles; when it is used up, the robot must act and the habit chooses. Budgets: busy, uniform 3–8; free, uniform 12–16. They were chosen from the measured cost of a deliberation on development seeds (9–11 cycles unpractised, 4 once chunked).

- **Chunking.** Each deliberation is compiled into a rule, so a practised conflict costs one decision instead of several.

Fairness of the contrast

M3 is a first draft (v0), designed by the first author to be a reasonable native Soar model, not a weak one. It will be revised with a student who works in Soar and, ideally, reviewed by a Soar expert before submission. Its known simplification: control knowledge, once derived, is applied deterministically (Soar's symbolic preferences override numeric ones). This is a property of Soar, not an added assumption, but alternative native designs (for example, RL over control operators) are possible and will be discussed.

4.4 Baselines

Plan only ($p_{\text{explicit}} = 1$) and *habit only* ($p_{\text{explicit}} = 0$) for M1. For M3: *habit only* (no deliberation) and *always deliberate* (deliberate on every cycle).

5 Simulation requirements

Any implementation of the task, in the grid world or in Webots, must meet these requirements so that the architectures are compared on equal terms.

1. **One world, one loop.** All models run in the same world code and episode loop (pyclarion-sc `ClarionCleaner.run`); only the decision function differs.
2. **Same interface.** The perception given to every model contains the same information: room, whether the room needs cleaning, quiet zone, the most attractive visible unassigned dirt and its path distance, battery, whether the assigned rooms are done, whether the robot is at the dock. A model may discretize it (M3) but may not receive more.
3. **Same timing.** One decision every three steps, or sooner if the behaviour cannot continue.
4. **Same costs and safety.** 0.1% battery for every explicit decision (top level in M1/M2, a deliberation in M3); the same 8% safety reflex outside the models.
5. **Same phases.** The scenario manipulations depend on the robot's phase of the task (cleaning A, approaching C's door, past C's door, inside C, cleaning B, returning). The phase is computed by the environment, the same way for every model.
6. **Logging.** Every decision records: time, phase, room, battery, temptation, chosen behaviour, level that chose it (top/bottom or controlled/habit), and the model's internal variables where they exist (drives, goal, $value(g)$, p_{explicit}; deliberation started, timed out).
7. **Reproducibility.** Fixed seeds; the seed sets the dirt layout, so conditions with the same seed share a world. Software: Python 3.13, NumPy, pyclarion-sc 0.5.0, Soar 9.6.5 (`soar-sml`). One command reproduces every table.

6 Measures

A decision is an **exposure** when unassigned dirt is visible ($temptation > 0$) and the robot is not inside room C. A **lapse** is an exposure at which the robot chooses `detour`.

Measure (per episode)	**Definition**
Primary	
Lapse rate before / after the door	lapses / exposures, while approaching C's door / after passing it
Lapse rate after room A	lapses / exposures in every phase after room A is clean
Lapse rate while busy	lapses / exposures during the map rebuild (end of A until B is clean)
Secondary	
Room C cleaned	% of room C's initial dirt removed
Task performance	steps to finish; episodes finished; battery used; quiet-zone violations
Deliberation cost	planning calls (explicit decisions or deliberations)
Manipulation checks	
M1/M2	mean p_{explicit} before vs. after the door; achievement deficit entering room B; p_{explicit} while busy
M3	deliberations started; deliberations cut short by the time budget

A measure is undefined (and the episode is left out of that test) when the episode has no exposure of the relevant kind. The number of episodes in every test is reported.

7 Hypotheses

M1, Clarion (confirmatory)

H1a In the *anticipation* condition, the lapse rate is higher after the door than before it.

H1b This before–after difference is larger in the *anticipation* than in the *constant* condition.

H2 The lapse rate after room A is higher in the *quiet* than in the *free* condition (motivational shift).

H3 The lapse rate while busy is higher with a *busy* than with a *free* computer.

M3, native Soar (confirmatory)

H4a No anticipation effect. The two anticipation conditions differ only in a drive stimulus, which M3 does not have, so the conditions are identical for M3 by construction. This is reported as a structural result, not as an empirical test.

H4b No motivational shift: the lapse rate after room A in the *quiet* condition is not higher than in the *free* condition (non-inferiority margin 0.05).

H4c No capacity effect with chunking: lapse rates while busy are equivalent between *busy* and *free* (equivalence margin ± 0.05), because practised control is cheap.

H4d Without chunking (ablation N1), the lapse rate while busy is higher with a *busy* computer.

M2 versus M1 (equivalence)

H5 In all six conditions, M2 is equivalent to M1 on the overall lapse rate, room C cleaned, steps and planning calls, with equivalence bounds of ±0.3 standard deviations of M1.

Exploratory (no confirmatory claim): H6, CLARION's position between the plan-only and habit-only baselines in lapses and planning calls; the parameter sweep (Section 9.1).

The hypotheses for M3 were informed by development runs (seeds 5001–5020). They are tested on the confirmatory seeds, which were not run.

8 Ablations

Each ablation removes one mechanism and states what should happen if that mechanism is responsible for an effect. All ablations run in all six conditions.

Ablation	What is removed	Prediction
A1 fixed drives	stimulus fixed at 0.65; no deficit decline ($k = 0$)	H1 and H2 effects disappear; H3 remains
A2 no cost	$c = 0$	net benefit always positive: high p_{explicit}, few lapses; effects shrink
A3 fixed p_{explicit} (0.25, 0.5, 0.75)	the MCS choice of explicitness	H1, H2 disappear; lapses fall as p_{explicit} rises
A4 random explicitness	drives and cost in the choice of level	H1, H2 disappear; lapses near A3 at 0.5
A5 no goal persistence	goal structure (goal re-drawn every cycle)	exploratory; little effect on H1–H3 expected
N1 no chunking (M3)	compilation of deliberation	a capacity effect appears (H4d)

9 Analysis plan

- **Sample.** 200 episodes (seeds 1–200) per model, variant and condition. The seed fixes the world, so conditions with the same seed are matched.
- **Power.** With 200 matched pairs, a paired test at $\alpha = .05$ (two-sided) has 80% power for a standardized effect of $d_z = 0.20$; equivalence tests at ±0.3 SD have at least 80% power when the true difference is zero.
- **Tests.** Between-condition hypotheses (H1b, H2, H3, H4b–d): paired t-tests on the per-seed difference, using seeds with the measure defined in both conditions. H1a: paired t-test within episodes. Directional hypotheses are supported only if the effect is significant *and* in the predicted direction. Robustness check: Wilcoxon signed-rank test.
- **Effect sizes.** Mean difference and d_z, with 95% bootstrap confidence intervals (10,000 resamples, percentile method).
- **Multiple comparisons.** Holm correction within each confirmatory family: {H1a, H1b, H2, H3} for M1 and {H4b, H4c, H4d} for M3. Manipulation checks and exploratory analyses are not corrected and are labelled as such.

- **Equivalence.** Two one-sided tests (TOST, paired) at $\alpha = .05$. H5 uses bounds of ± 0.3 SD of M1 for each measure; equivalence for H5 requires all 24 tests to pass. H4b (non-inferiority) and H4c use absolute margins on the lapse rate (0.05), because M3's lapse rate can be constant, which makes SD-based bounds undefined.
- **Exclusions.** None, other than undefined measures. Episodes that hit the step limit are kept.

9.1 Robustness (exploratory)

For M1, the H1b and H2 effects are recomputed on a grid of the two free MCS parameters, cost $c \in \{0.12, 0.16, 0.185, 0.22, 0.26\}$ and temperature $\tau \in \{0.4, 0.7, 1.0\}$, with 100 seeds per cell. The question is whether the direction of each effect holds across a region of parameter space, not only at the published values (Pitt et al., 2006).

10 Webots and the physical robot

The grid world is the main experimental setting: it is fast, controlled and gives enough power. Webots and the physical robot test whether the effects survive embodiment (continuous motion, sensor noise, real navigation). They are reported in Document 4.

- **Simulator.** Webots R2025a, the current stable release (February 2025; later builds are nightly).
- **Robot.** The e-puck (Webots' e-puck model, version 2), as the student proposed. It has 8 infrared proximity sensors, a camera, an IMU and two stepper motors, which are enough for this task.
- **Arena.** The grid world scaled so that one cell is about the size of the robot (7 cm), about 1.9×0.7 m, with the same rooms, doors and dock. Dirt is a virtual layer kept by the controller from the robot's position.
- **Same brains.** Each controller calls the same decision code as in the grid world (M1 through pyclarion-sc, M3 through Soar); only perception and motion change.
- **Physical e-puck2.** The decision code runs on a laptop and commands the robot over Wi-Fi or Bluetooth. Position comes from an overhead camera with fiducial markers, since there is no GPS. The physical runs are a demonstration (about 10 runs per condition), not a confirmatory test. Before purchase, remote control of a real e-puck2 from our Webots version must be checked.

11 Threats to validity

- **A weak native model.** If M3 is a poor Soar model, the contrast says little about Soar. The model will be revised with a Soar user and the design choices are listed in Document 3.
- **Chosen parameters.** The robot's world quantities, M3's discretization and its time budgets are chosen, not fitted. The parameter sweep and the ablations show how much the conclusions depend on them.

- **Built-in effects.** Some effects follow directly from how a manipulation reaches a model (for example, H4a). These are reported as structural, not as discoveries.
- **From robot to people.** The robot results test the mechanisms, not human behaviour. The link to human data comes from the earlier simulations.

12 Deviations and versioning

This document is version 1.0. Any change after the confirmatory runs (to models, measures, tests or exclusions) will be listed in Document 4 with its reason, and analyses affected by it will be labelled exploratory. Code, data and this protocol will be released with a DOI on publication.

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